

HW # 16

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CS-F

Problem 1:

$$y = mx \Rightarrow y = \begin{bmatrix} 1 \\ m \end{bmatrix} x$$

$$u = \frac{1}{\sqrt{1+m^2}} \begin{bmatrix} 1 \\ m \end{bmatrix}$$

$$uu^T = \frac{1}{m^2+1} \begin{bmatrix} 1 \\ m \end{bmatrix} \begin{bmatrix} 1 & m \end{bmatrix}$$

$$2uu^T = \frac{1}{m^2+1} \begin{bmatrix} 2 & 2m \\ 2m & 2m^2 \end{bmatrix}$$

$$2uu^T - I = \begin{bmatrix} 2/m^2+1 - 1 & 2m/m^2+1 \\ 2m/m^2+1 & 2m^2/m^2+1 - 1 \end{bmatrix}$$

$$= \frac{1}{m^2+1} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}$$

Proved:-

Problem 2:

$$R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

rotate by $-\theta$

$$R_{-\theta} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

reflect across x-axis:

$$R_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

again rotate by θ :

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

final matrix:

$$\begin{aligned} F &= R_\theta R_x R_{-\theta} \\ &= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos 2\theta & 2\cos\theta \sin\theta \\ 2\cos\theta \sin\theta & -\cos 2\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \end{aligned}$$

Problem 3:

$$\begin{aligned} \text{a) } F_v &= 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= 2 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ F_v &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \end{aligned}$$

$$b) \begin{cases} y = 1x \\ y = mx \end{cases}$$

$$F_L = \frac{1}{2} \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$c) F_0 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Problem 4:

$$\begin{aligned} A \cdot A &= U U^T U U^T \\ &= U (U^T U) U^T \quad \text{since } (U^T U) = I \\ &= U U^T \end{aligned}$$

$$A^2 = A$$

It projects vectors on line spanned by unit vector U . Applying more than once has no effect since it is already on U . That's why it is projection in $\mathbb{R}^n$.

Problem 5:

$$\begin{aligned} a) A &= U U^T \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \end{aligned}$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{Projects on x-axis.}$$

b) w is not a unit vector.

$$u = \frac{1}{\sqrt{14}} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{14} \\ 2/\sqrt{14} \\ 3/\sqrt{14} \end{bmatrix}$$

$$\begin{aligned} A &= uu^T \\ &= \begin{bmatrix} 1/\sqrt{14} \\ 2/\sqrt{14} \\ 3/\sqrt{14} \end{bmatrix} \begin{bmatrix} 1/\sqrt{14} & 2/\sqrt{14} & 3/\sqrt{14} \end{bmatrix} \\ &= \begin{bmatrix} 1/14 & 2/14 & 3/14 \\ 2/14 & 4/14 & 6/14 \\ 3/14 & 6/14 & 9/14 \end{bmatrix} \end{aligned}$$

$$A = \begin{bmatrix} 1/14 & 1/7 & 3/14 \\ 1/7 & 2/7 & 3/7 \\ 3/14 & 3/7 & 9/14 \end{bmatrix}$$

Projects w on unit vector u
 u in the same direction as w .